

Proof. Let

$$\mathbf{a} = (a_1, \dots, a_n), \quad \mathbf{b} = (b_1, \dots, b_n) \in \mathbb{R}^n.$$

The Euclidean inner product and norm are

$$\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{i=1}^n a_i b_i, \quad \|\mathbf{a}\|^2 = \sum_{i=1}^n a_i^2, \quad \|\mathbf{b}\|^2 = \sum_{i=1}^n b_i^2.$$

We must prove $\langle \mathbf{a}, \mathbf{b} \rangle^2 \leq \|\mathbf{a}\|^2 \|\mathbf{b}\|^2$.

A non-negative quadratic form. For an arbitrary real parameter t consider

$$Q(t) = \sum_{i=1}^n (a_i - t b_i)^2 \geq 0 \quad (t \in \mathbb{R}).$$

Expansion of $Q(t)$. Using $(a - tb)^2 = a^2 - 2tab + t^2b^2$,

$$\begin{aligned} Q(t) &= \sum_{i=1}^n (a_i^2 - 2ta_i b_i + t^2 b_i^2) \\ &= \left(\sum_{i=1}^n b_i^2 \right) t^2 - 2 \left(\sum_{i=1}^n a_i b_i \right) t + \left(\sum_{i=1}^n a_i^2 \right). \end{aligned}$$

Set $q = \sum b_i^2$, $p = -2 \sum a_i b_i$ and $r = \sum a_i^2$.

Discriminant argument. If $q = 0$ then $\mathbf{b} = \mathbf{0}$ and the inequality is trivial. Assume $q > 0$. Since $Q(t) \geq 0$ for all t , the discriminant of the quadratic $qt^2 + pt + r$ must satisfy $p^2 - 4qr \leq 0$; that is,

$$\left(-2 \sum a_i b_i \right)^2 - 4 \left(\sum b_i^2 \right) \left(\sum a_i^2 \right) \leq 0.$$

Dividing by $4 > 0$ yields

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right),$$

which is the Cauchy–Schwarz inequality.

Equality case. Equality holds exactly when the discriminant is zero, i.e. when the quadratic has a double root. Then $Q(t) = q(t - t_0)^2$ for some $t_0 \in \mathbb{R}$, and $Q(t_0) = 0$ forces $a_i = t_0 b_i$ for every i . Hence there exists $\lambda = t_0$ such that

$$a_i = \lambda b_i \quad \text{for all } i,$$

or one of the vectors is the zero vector. Thus equality occurs precisely when the two vectors are linearly dependent. □